4Developers Katowice

Refactoring
using Scatter-gather

Cezary Sanecki
Allegro

Zaobserwuj 4Developers i bądź zawsze na bieżąco!

Software Engineer @ Allegro
Letting people get to know me
Ex-civil engineer
WJUG leader

CEZARY SANECKI

AGENDA

DOMAIN

AS-IS STATE

BUSINESS CHANGE

FINDING SOLUTION

OFFTOP

IMPLEMENTATION & PROBLEMS

SUMMARY

DOMAIN?

ADVERTISING IN ZX

CREATE ACCOUNT FOR USER

EMIT ADS

POLL STATS

POLL STATS

RESPONSIBILITES

poll
statistics
(obviously!)

join
two
worlds

publish
aggregated
stats

AS-IS PICTURE

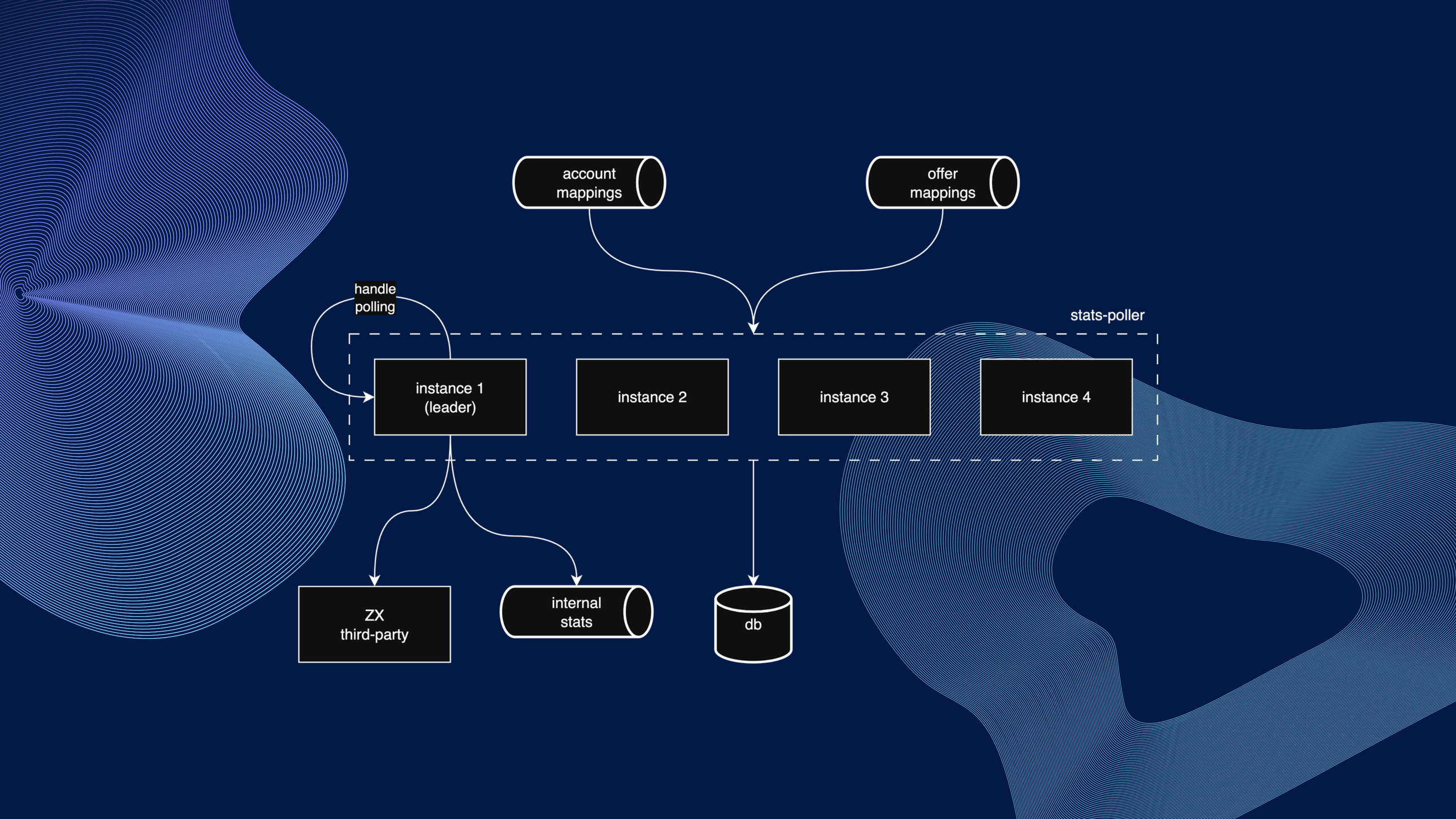
6
instances
per DC

25 min
polling for all
accounts

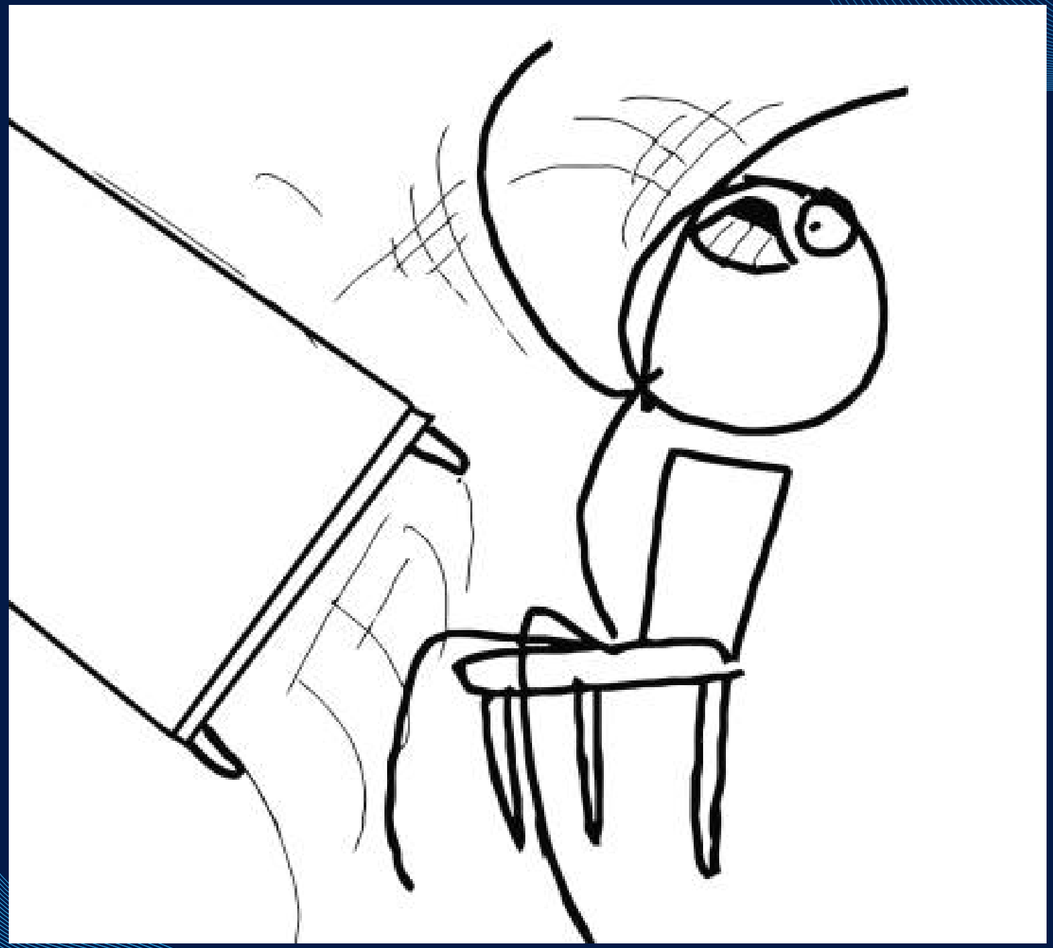
reactive programming

IMPORTANT MECHANISMS

- leader poll stats
- after polling marks it is done
- all instances handle mappings
- metrics and alerts


account mappings
offer mappings
handle polling
stats-poller
instance 1 (leader)
instance 2
instance 3
instance 4
ZX third-party
internal stats
db

BUSINESS ARRIVES...



VISION

more accounts to
handle, like ~10x

BRAIN STORMING

BRAIN STORMING

more
instances

other
instances
leader

BRAIN STORMING

more instances

better threads usage

TOO MUCH WORK...
AND UNCERTAIN RESULT

BRAIN STORMING

more instances

better threads usage

split responsibilites

FOUND RESPONSIBILITIES

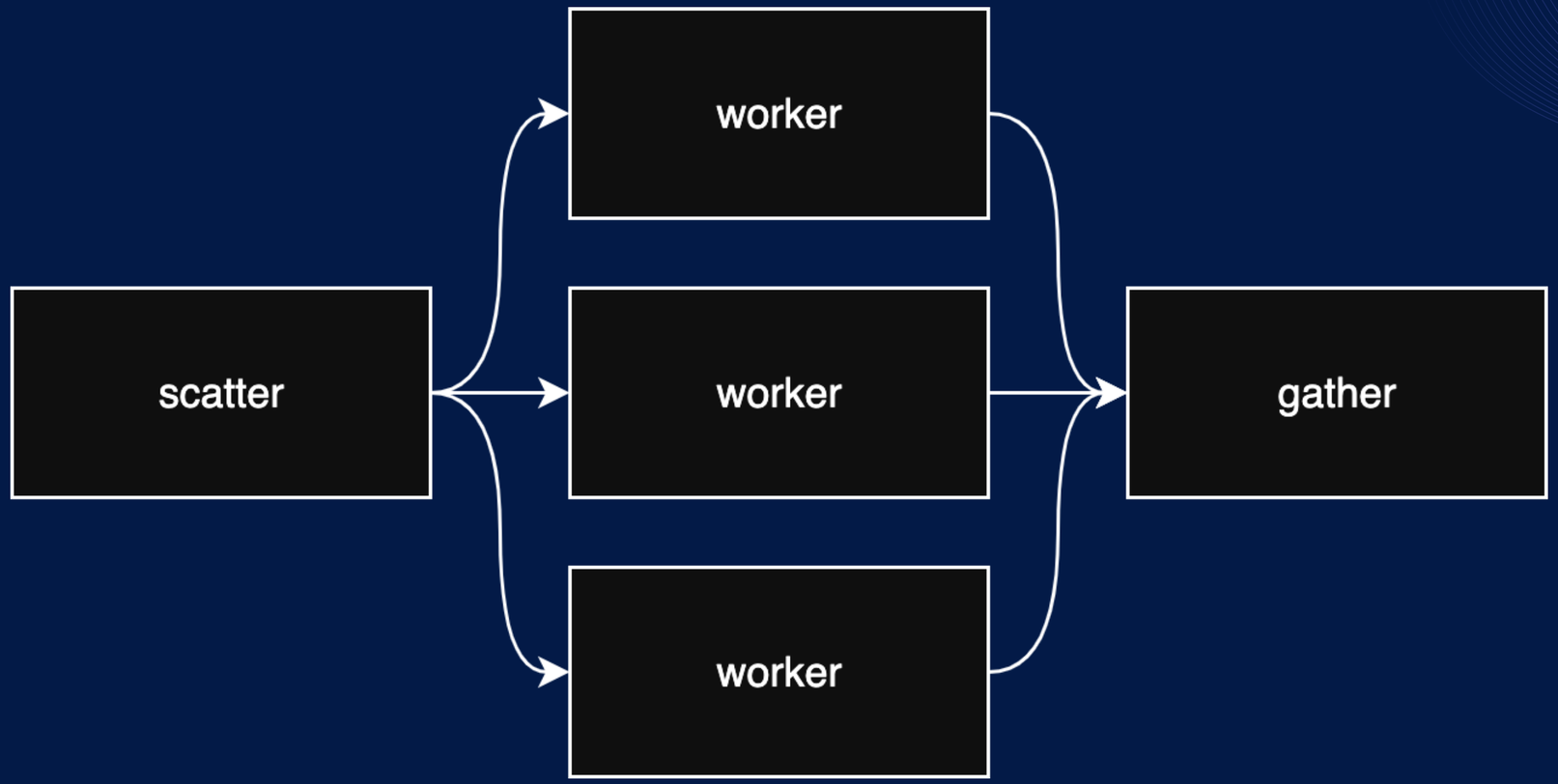
trigger

polling

polling

polling

FINDING THIS IS...
A LOT OF WORK!


worker
scatter
worker
gather
worker

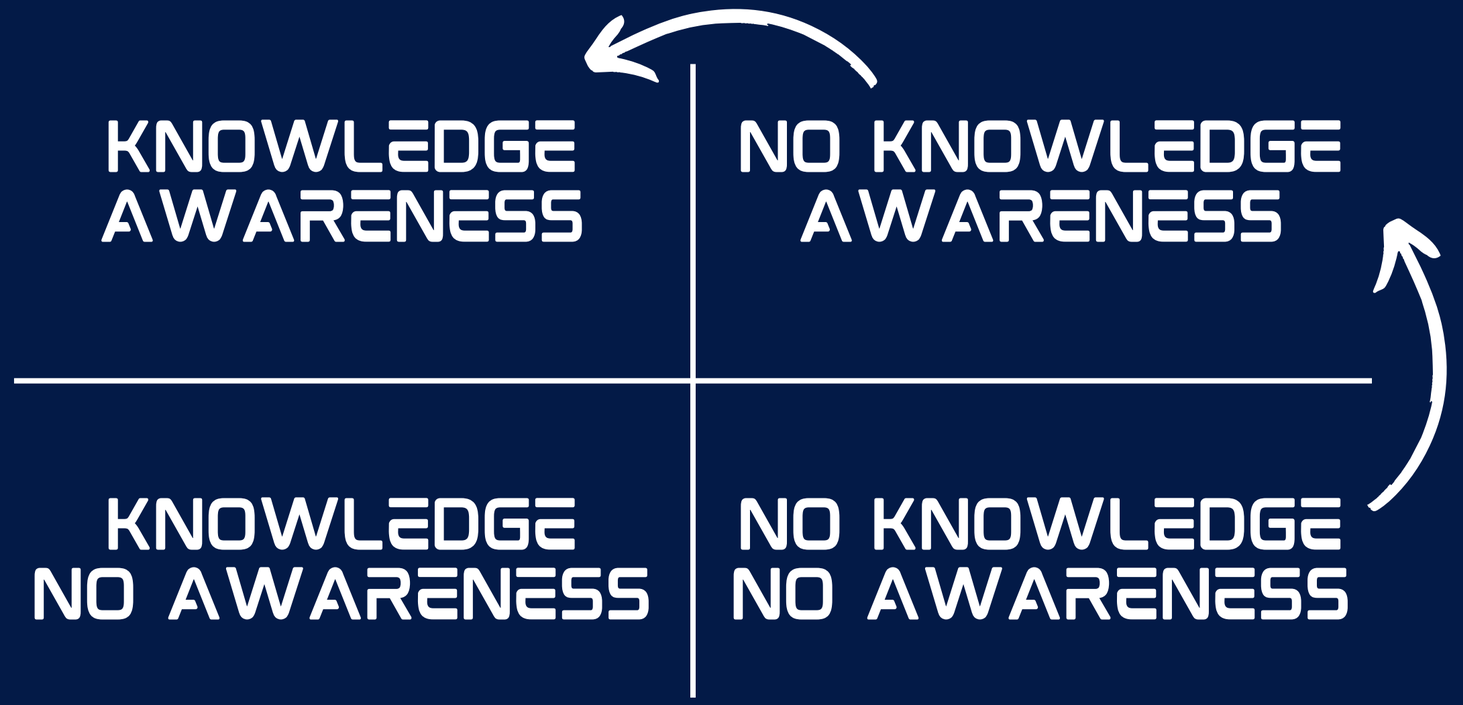
SCATTER GATHER

HOW DO THAT?

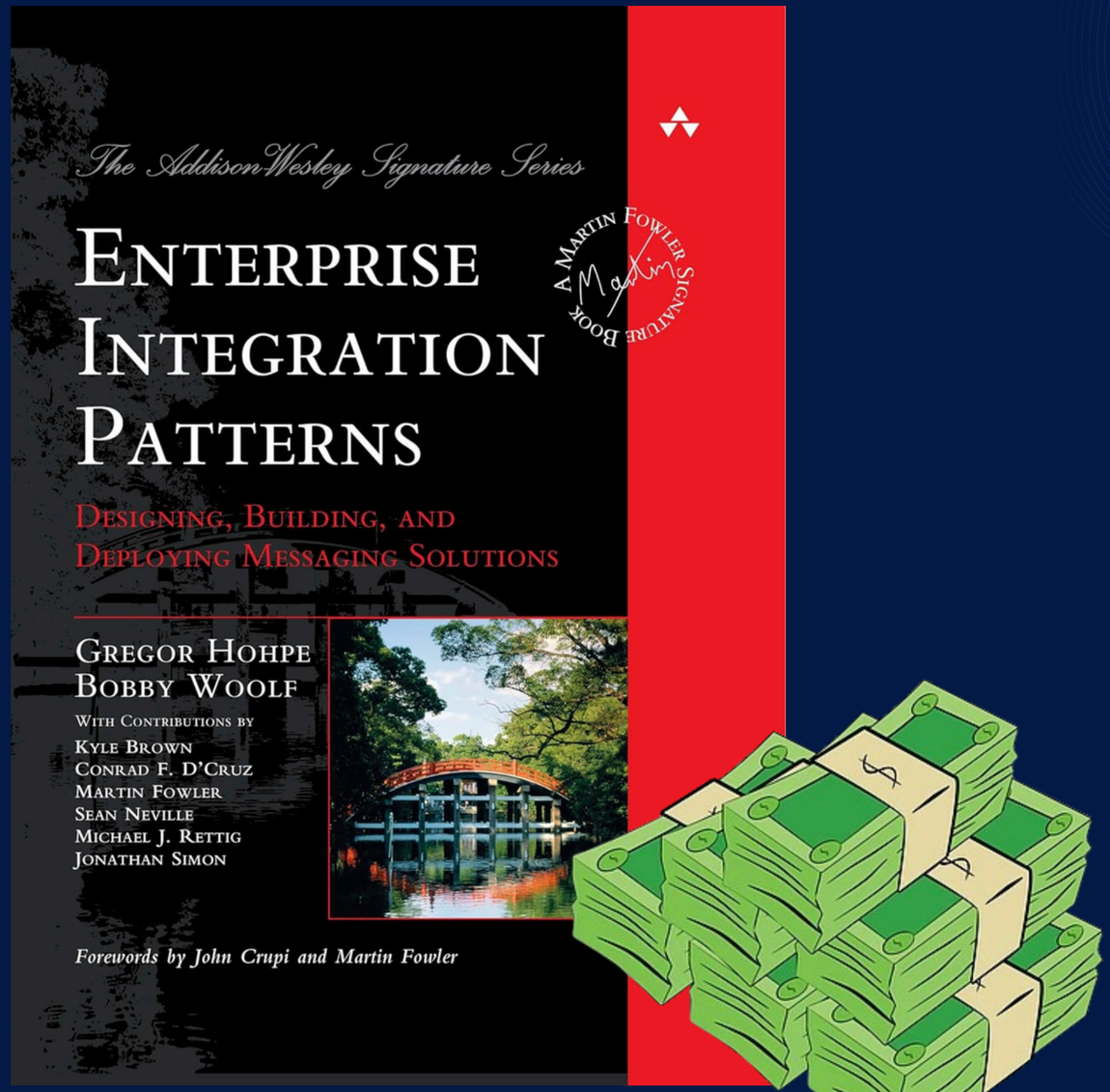
RUMSFELD MATRIX

KNOWN KNOWNS	KNOWN UNKNOWNS
UNKNOWN KNOWNS	UNKNOWN UNKNOWNS

RUMSFELD MATRIX



WHERE TO GET KNOWLEDGE?


The Addison Wesley Signature Series
A Martin Fowler Signature Book
Enterprise Integration Patterns
Designing, Building, and Deploying Messaging Solutions
Gregor Hohpe
Bobby Woolf
With Contributions by
Kyle Brown
Conrad F. D'Cruz
Martin Fowler
Sean Neville
Michael J. Rettig
Jonathan Simon
Forewords by John Crupi and Martin Fowler

WE NEED TO GO
CHEAPER

Better
Software
Design

Better Software Design

ODC. 96

O DOSTARCZANIU EVENTÓW W SYSTEMACH ROZPROSZONYCH Z MICHAŁEM OSTRUSZKĄ

LISTEN TO YOURSELF

LISTEN TO YOURSELF

QUICK PROCESSING

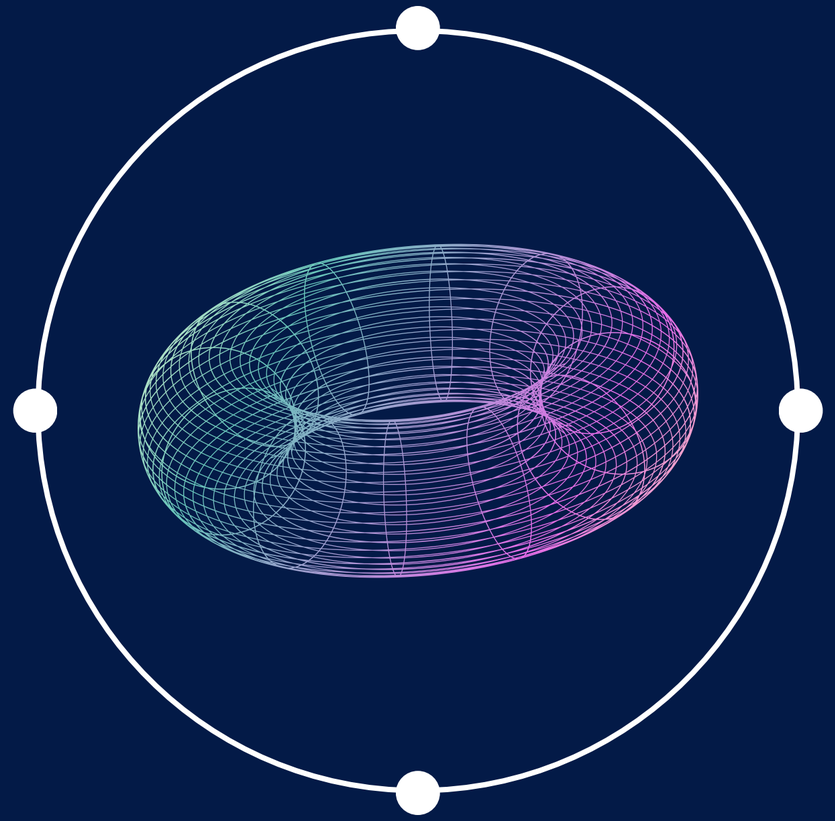
EVENTUAL CONSISTENCY

RARELY APPLIED

COMMAND VS EVENT

HOW TO IMPLEMENT SCATTER GATHER?

1. split responsibilites

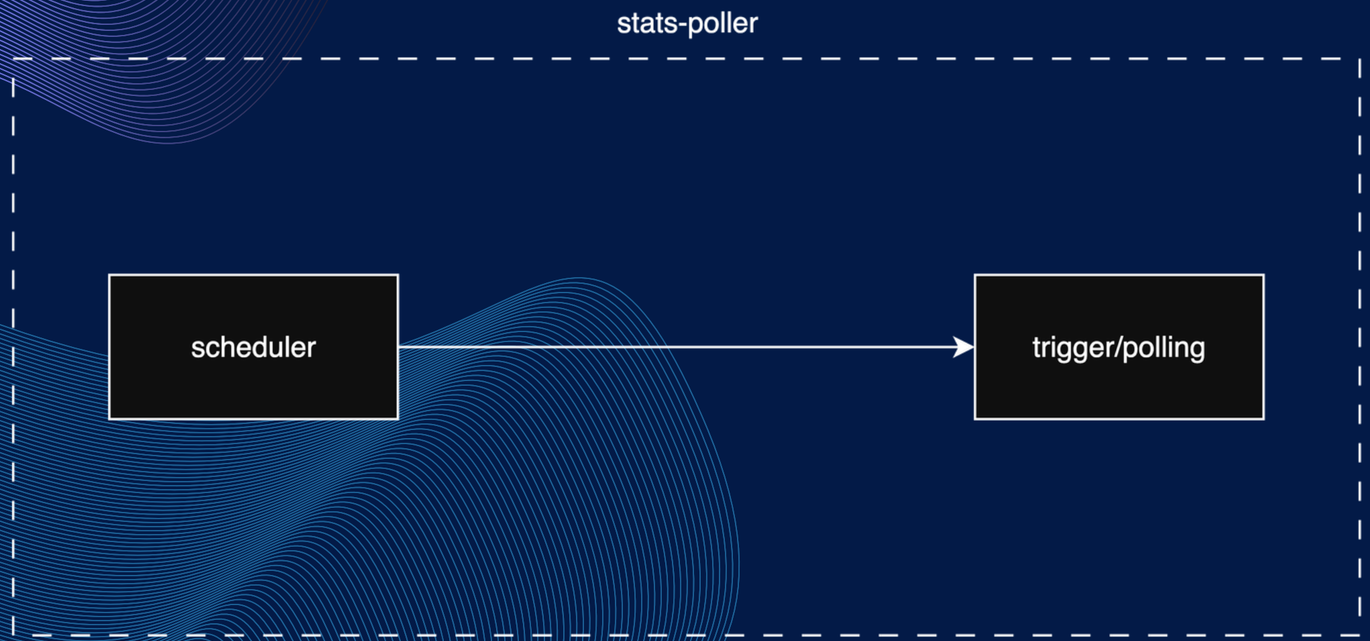


2. provide abstraction

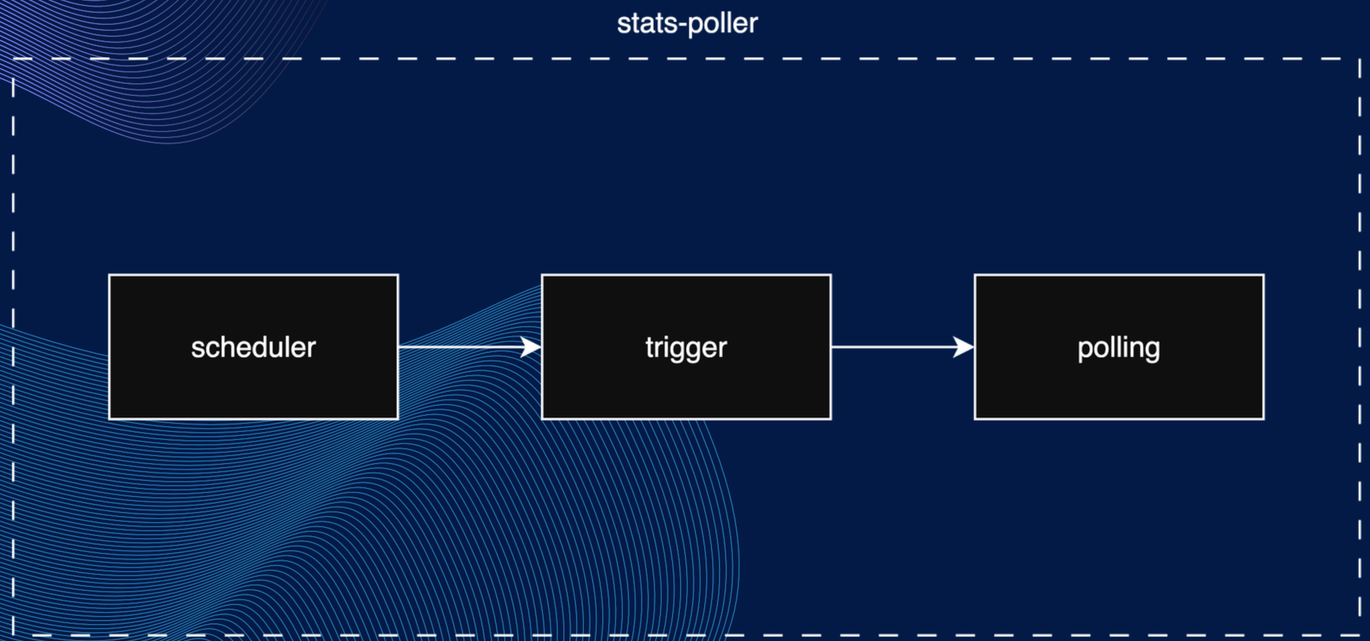
3. create “adapter”

4. switch solution

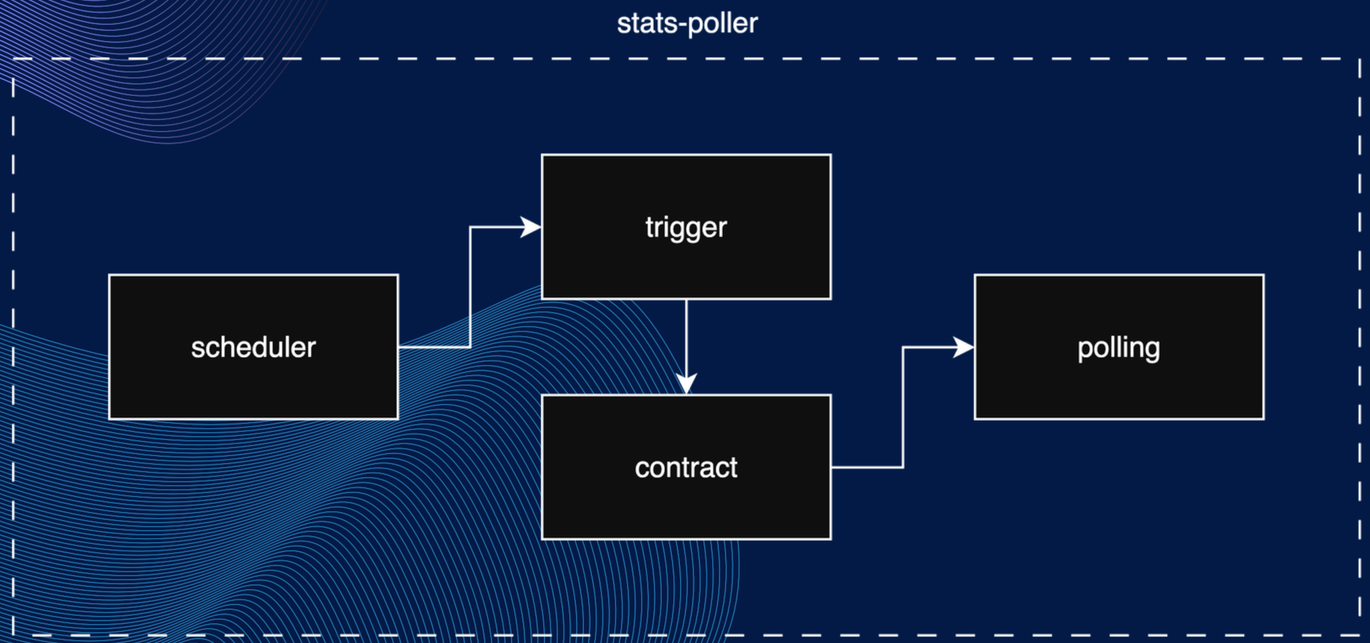
CURRENT STATE



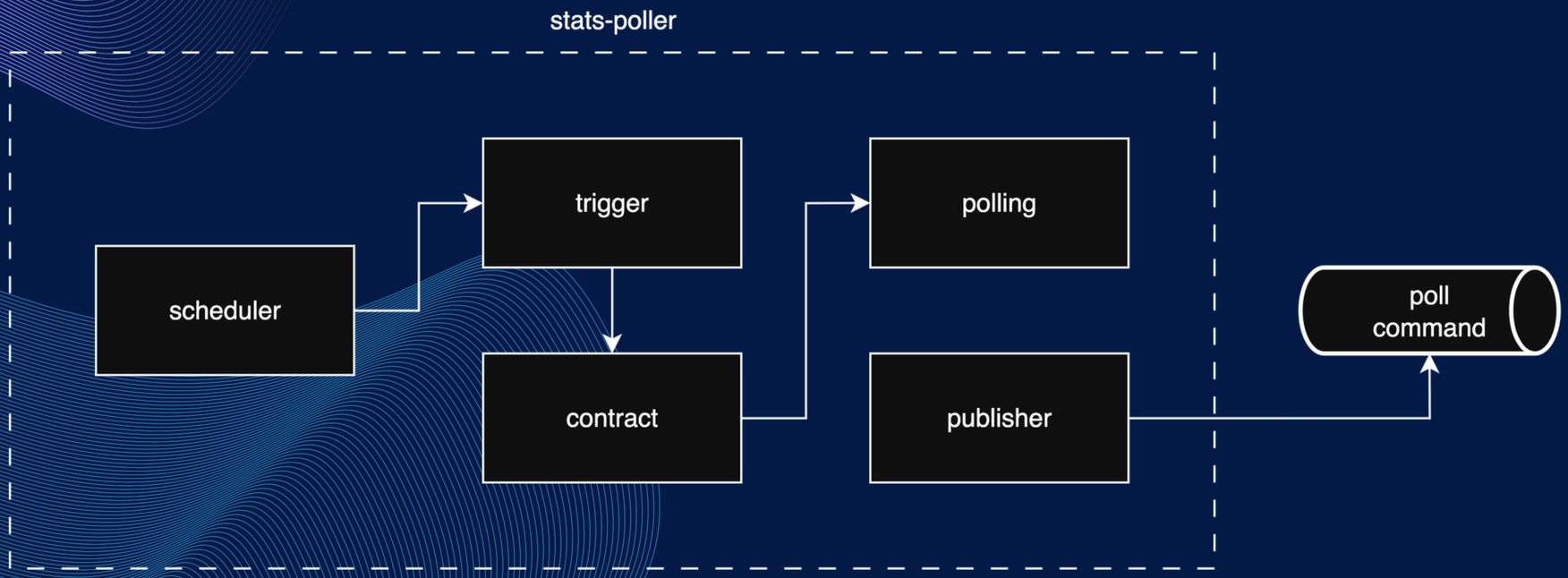
SPLIT RESPONSIBILITES



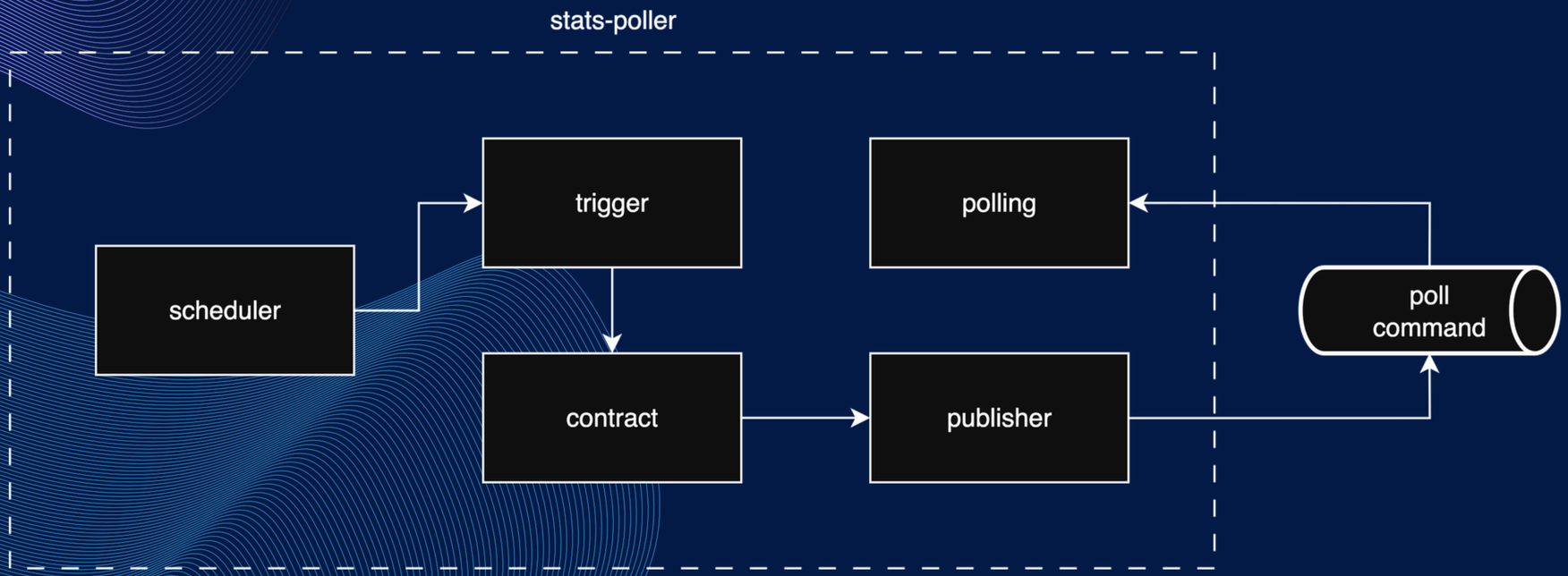
PROVIDE ABSTRACTION



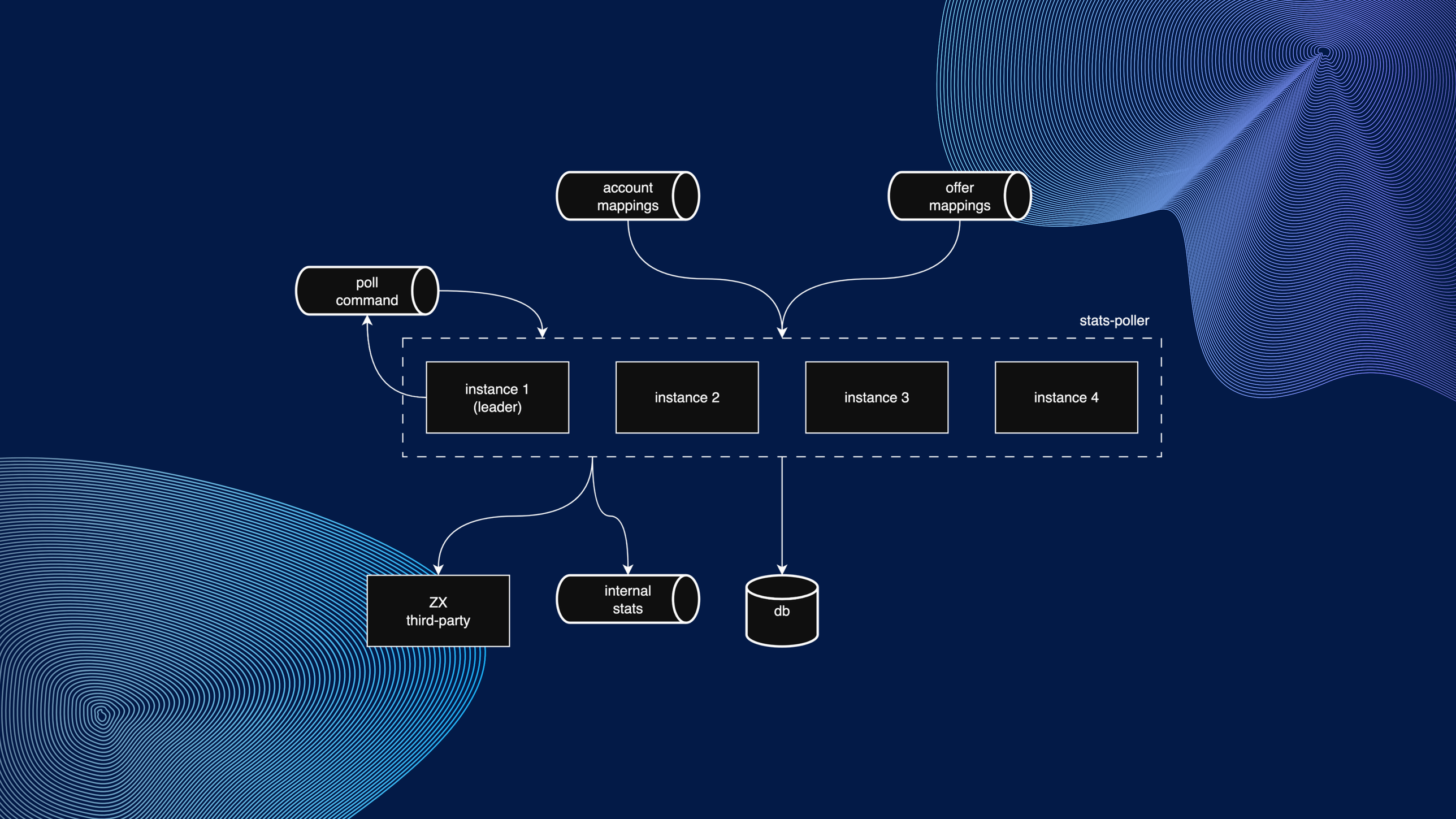
CREATE PUBLISHER



SWITCH SOLUTION

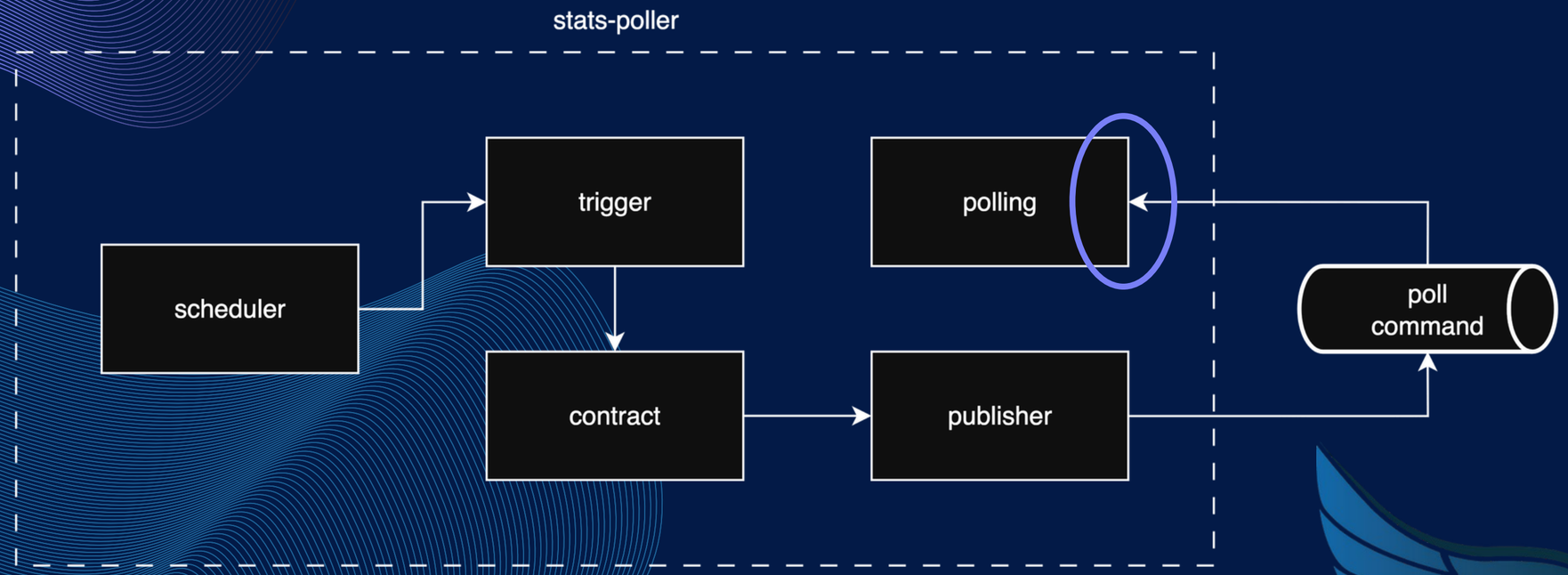


OVERVIEW


account mappings
offer mappings
poll command
stats-poller
instance 1 (leader)
instance 2
instance 3
instance 4
ZX third-party
internal stats
db

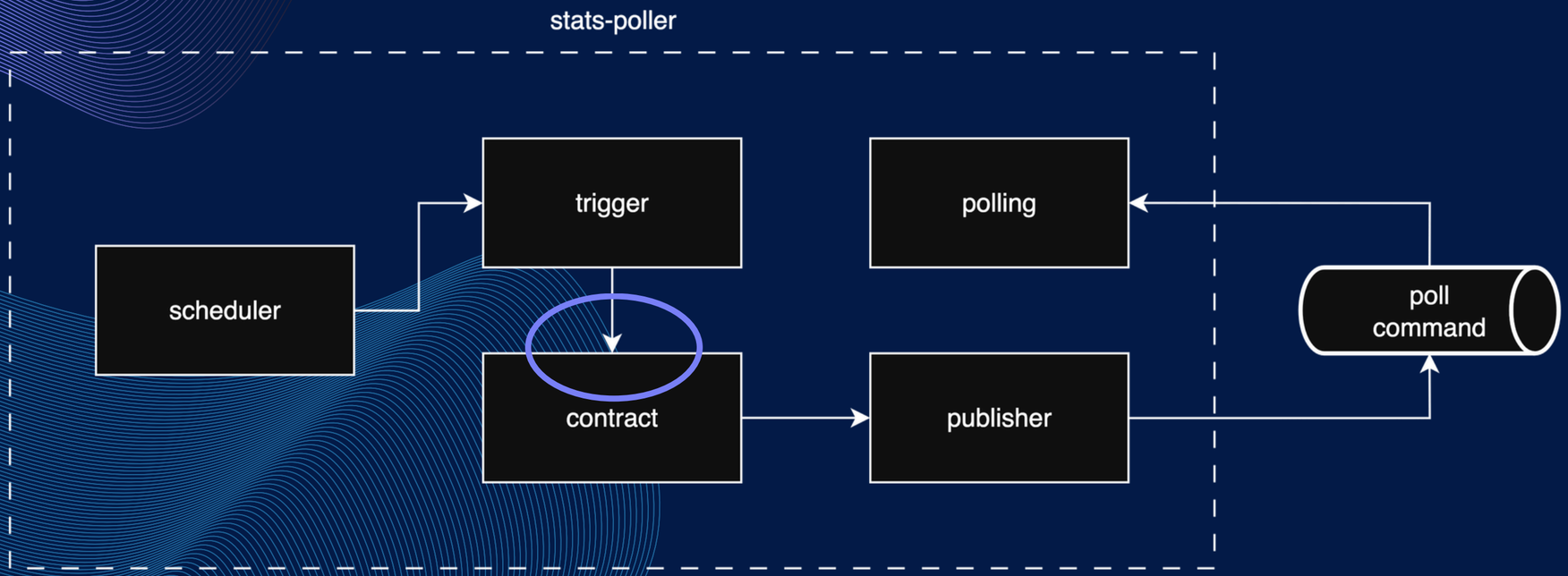
PROBLEMS?
OF COURSE!
I HAVE MANY

TIMEOUT ON HANDLER

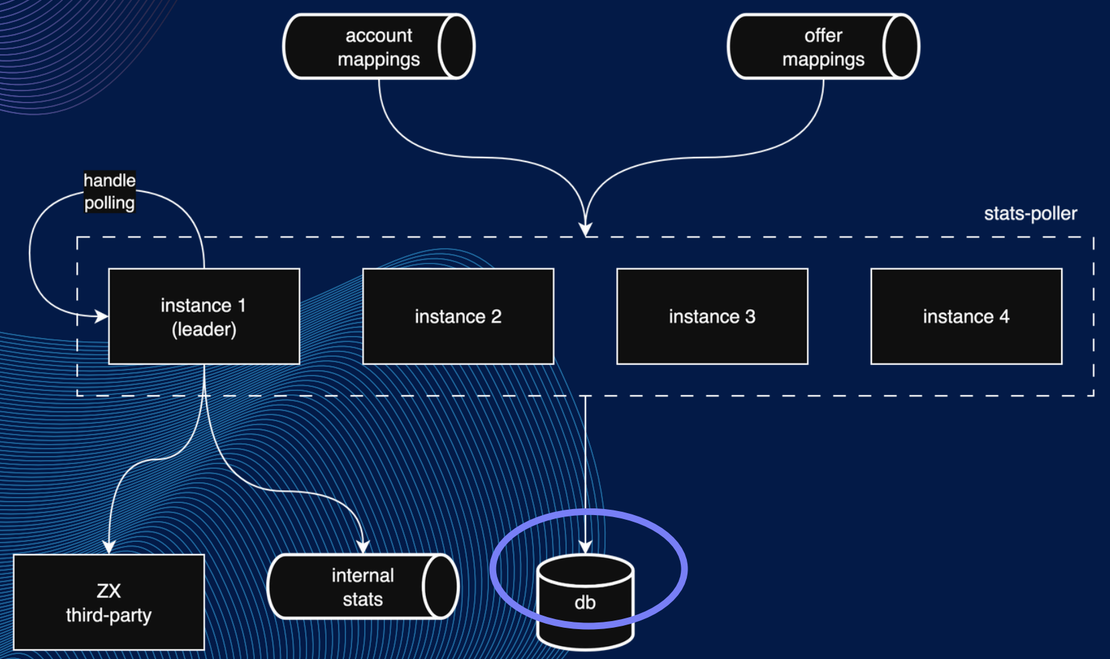


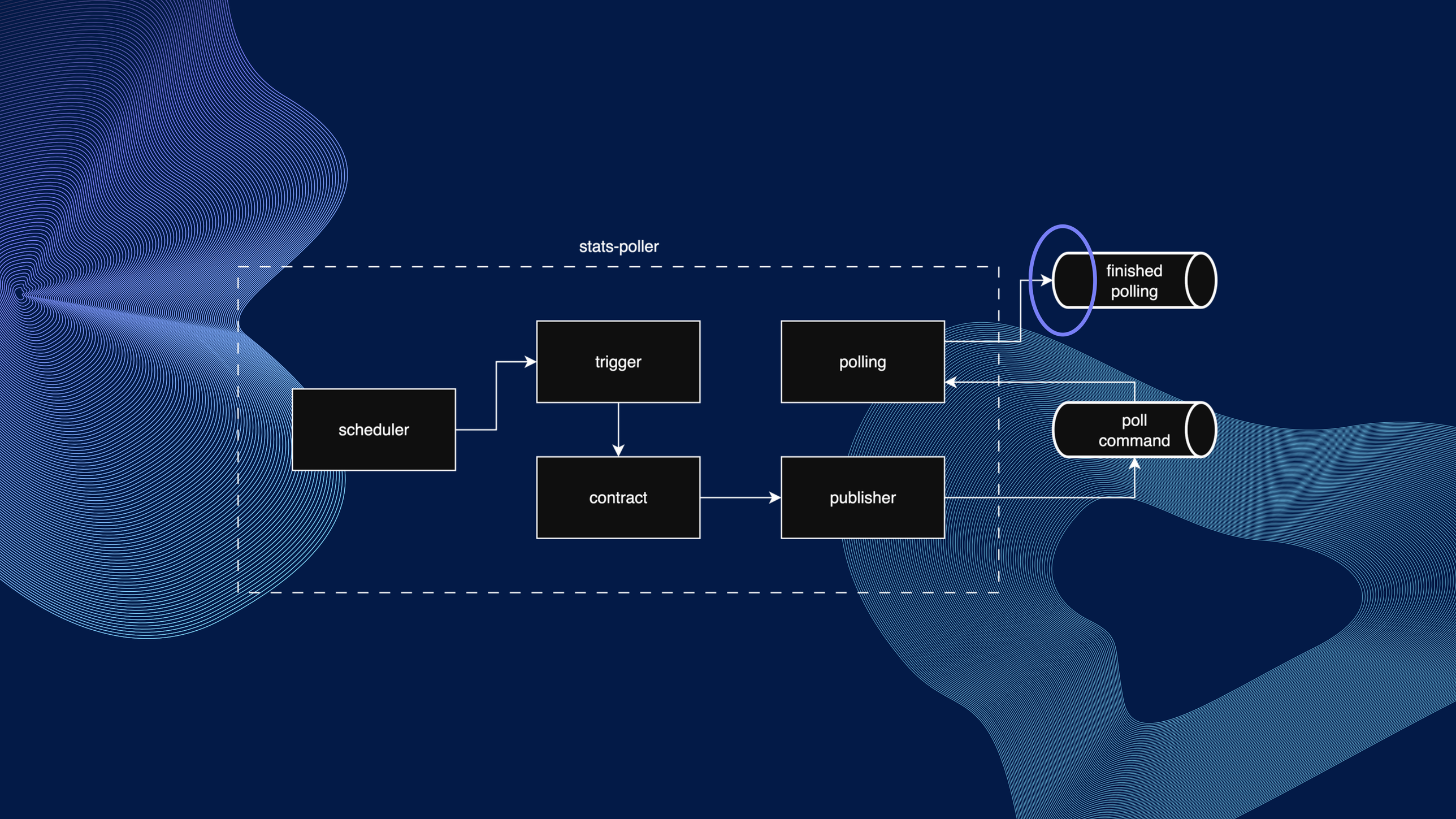
HERMES

BUSINESS UNAWARENESS

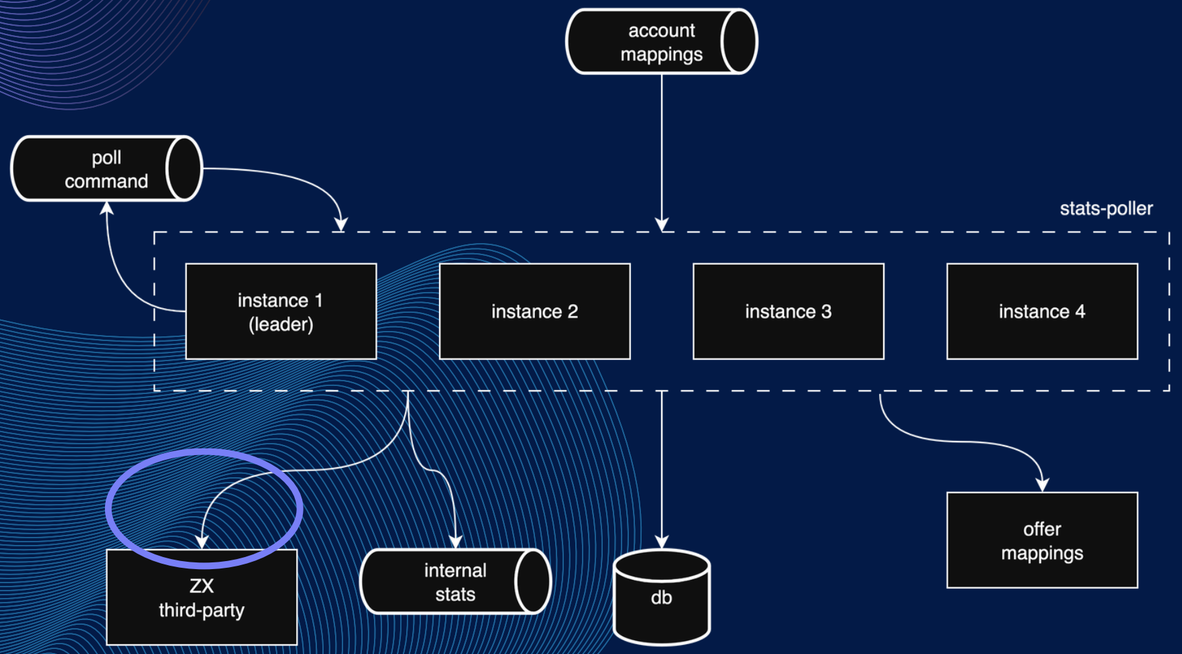


FINISH MARKERS




stats-poller
scheduler
trigger
contract
polling
publisher
finished
polling
poll
command

THIRD PARTY LIMITS



OUR AIM IS TO HANDLE MORE ACCOUNTS, BUT...

CURRENT STATE

3
instances
per DC

8 min
polling for all
accounts

more
readable code

WE WILL SEE WHAT FUTURE WILL BRING...

THANK YOU!

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Web: cezarysanecki.pl
GitHub: cezarysanecki

developers
KATOWICE

Dziękuję

Oceń mój wykład w aplikacji Eventory

eventory

1. Wejdź w agendę
2. Kliknij w wybrany wykład
3. Oceń i zostaw komentarz